

Marc DAMIE

Applied Cryptographer, Ph.D.

<https://marc.damie.eu/> | [GitHub](#) | [LinkedIn](#) | [CryptoHack](#) | [Mastodon](#) | [GScholar](#)

Education

Ph.D. in Computer Science

Apr. 2022 - Dec. 2025

Université de Lille

Title: Privacy-Preserving Computations on Sparse Data.

Supervisors: Jan Ramon (Inria, FR), Andreas Peter (University of Oldenburg, DE), and Florian Hahn (University of Twente, NL).

Master's Degree in Computer Science (i.e., "*Diplôme d'ingénieur*")

Sep. 2016 - Aug. 2021

Université de Technologie de Compiègne

Major: Data Mining. **3 Minors:** Philosophy / Technological Controversies / Math Modelling. **GPA:** 5/5.

Employment

Radboud University

Sep. 2026 -

Scientific Programmer on the Authentimark project (within iHub)

University of Twente

Jan. 2026 - May 2026

Postdoctoral researcher in the Semantics, Cybersecurity and Services group

University of Twente

Oct. 2023 - Dec. 2025

Doctoral researcher in the Semantics, Cybersecurity and Services group

Inria

Apr. 2022 - Sep. 2023

Doctoral contract in the MAGNET (MACHINE learning in information NETWORKs) team

Inria

Feb. 2021 - Mar. 2022

Research engineer in the MAGNET (MACHINE learning in information NETWORKs) team

University of Twente

Feb. 2020 - Jul. 2020

Research intern in the Semantics, Cybersecurity and Services group

Gatewatcher

Aug. 2019 - Feb. 2020

Back-end intern on an intrusion detection system

Volunteering

European Data Protection Board

2026-

Expert in the Support Pool of Expert reserve list

ctrl+alt+reclaim

2024-

Co-founder of a European collective promoting youth-driven digital regulations

Twente Hacking Squad

2023 - 2026

Member and active player in CTF challenges

Rhizome (non-profit Internet Service Provider)

2018 - 2022

Member then treasurer

Awards and Honors

July 2025

Distinguished Reviewer PETS AE 2025

March 2025

Young Talents in Cybersecurity 2025 (by the French Embassy in the NL and Dutch Embassy in FR)

May 2022

Nomination for Dutch Cyber Security best Research Paper (DCSRP) 2021

Publications

- [1] **M. Damie**, F. Hahn, and A. Peter, "A Highly Accurate Query-Recovery Attack against Searchable Encryption using Non-Indexed Documents", USENIX Security 2021
- [2] M. Dijkslag, **M. Damie**, F. Hahn, and A. Peter, "Passive query-recovery attack against secure conjunctive keyword search schemes", ACNS 2022
- [3] **M. Damie**, J-B. Leger, F. Hahn, and A. Peter, "Revisiting the attacker's knowledge in inference attacks against Searchable Symmetric Encryption", ACNS 2025
- [4] B. van Dartel, **M. Damie**, and F. Hahn, "Evaluating Membership Inference Attacks in heterogeneous-data setups", SiMLA/ACNS 2025
- [5] **M. Damie**, F. Hahn, A. Peter, and J. Ramon, "Eliminating Exponential Key Growth in PRG-based Distributed Point Functions", DPM/ESORICS 2025
- [6] **M. Damie**, F. Hahn, A. Peter, and J. Ramon, "DDH-based schemes for multi-party Function Secret Sharing", NordSec 2025
- [7] **M. Damie**, M. Pop, and M. Posthuma, "Energy Consumption of TLS, Searchable Encryption, and FHE", FPS 2025
- [8] **M. Damie**, F. Mazzone, F. Hahn, A. Peter, and J. Ramon, "Noisy Function Secret Sharing and its application to DP", 2025
- [9] **M. Damie**, F. Hahn, A. Peter, and J. Ramon, "How to securely shuffle? A survey about secure shufflers in privacy-preserving computations", 2025
- [10] **M. Damie**, and E. Cyffers, "Fedivertex: a graph dataset based on Decentralized Social Networks", WWW 2026
- [11] **M. Damie**, F. Hahn, A. Peter, and J. Ramon, "Secure Sparse Matrix Multiplications and their applications to PPML", CODASPY 2026
- [12] C. Gupta, N. Guldali, and **M. Damie**, "IoTLS 2.0: How Far Has IoT Industry Come in Securing Communications with TLS?", EuroSec/EuroSys 2026
- [13] **M. Damie**, B. Ertan, D. Essoussi, A. Makhanu, G. Peter, and R. Wensveen, "TOSSS: a CVE-based Software Security Benchmark for Large Language Models", IWSPA 2026
- [14] R. Ritz, **M. Damie**, T. van Ede, and F. Marzani, "A Robust and Privacy Preserving Crowd Monitoring Framework", ARES 2026
- [15] **M. Damie**, "SparseMPC: Secure Sparse Operations using Multi-Party Computation", 2026

Teaching

Supervision

2022-now

BSc: N. Guldali, G. Kuindersma, M. Pop, M. Posthuma, A. Verhovetchi, B. Sanowski

University of Twente

MSc: M. Dijkslag, M. van Dartel

Secure Data Management

2024-2026

Instructor

University of Twente

Architecture of Operating Systems

2023

Instructor

University of Lille

Programming 101

2020

Design of MOOC-like content

Freelance

System Administration

2018

Design of MOOC-like content

Student Association

Academic Service

Reviews

2022 - now

IEEE TDSC, IEEE TIFS, DIMVA 2026, UAI 2025, CCS 2024, PETS 2023, CODASPY 2022.

Artifact Evaluation

2023 - now

Member of the AE committees of PETS 2025, NDSS 2025, NDSS 2024, PETS 2024.

Scientific events

2024 - now

Part of the local organization of the CSF symposium 2024

Student Representative in the Board of Directors

2017 - 2021

Université de Technologie de Compiègne

Public Speaking

PublicSpaces Conference

Lightning talk on "A manifesto for a public service social media"

2026
Amsterdam, NL

"Moi et les autres" Festival

Roundtable on "Making social media a safe space"

2026
Paris, FR

Forum InCyber

Talk on "Is Vibe Coding your Enemy?"

2026
Lille, FR

Trust and Safety Forum

Roundtable on the regulation to protect minors online

2026
Lille, FR

ONE Conference

Talk on "Are you sure that your LLM is not poisoned?" (Young Talents programme)

2025
The Hague, NL

Silbersalz Festival

Round table on "Privacy in the digital world, mission impossible?"

2024
Halle, DE

ONE Conference

Round table on "Teaching ethics to (future) IT professionals"

2025
The Hague, NL

PublicSpaces Conference

Lightning talk on "Federated Machine Learning for the Fediverse"

2024
Amsterdam, NL

Media Coverage

Le Monde

Op-ed about the influence of digital technologies on GenZ.

23/09/2026

EUobserver

Reaction about the final vote of the social media ban for under-15s in France.

22/07/2026

El Periódico

Reaction about the vote of the social media ban for under-15s in the French Senate.

01/04/2026

Deutsche Welle

Interview about the social media ban for under-15s.

06/02/2026

France Info

Interview about the social media ban for under-15s.

26/01/2026

Le Monde

Op-ed about the social media ban for under-15s.

25/01/2026

Logging Off: The Human Cost of Our Digital World

Interview about the alternative digital spaces by Adele Zeynep Walton for her book.

2025

Science Communication

Silbersalz Institute

Training programme for scientists wanting to break into the media sphere.

2024

Scientific Game Jams

Co-created two small games presenting scientific concepts. My itch.io: MarcTOK

2024

Wikipedia contributions

More than 1000 contributions. Mostly on French-speaking Wikipedia.

2023 - now

Languages

French (Native), **English** (Professional/C1), **Dutch** (Intermediate/B1), **Spanish** (Beginner)